

A Survey on Trustworthy LLM Agents: Threats and Countermeasures

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Abstract

With the rapid evolution of Large Language Models (LLMs), LLM-based agents and Multi-agent Systems (MAS) have significantly expanded the capabilities of LLM ecosystems. This evolution stems from empowering LLMs with additional modules such as memory, tools, environment, and even other agents. However, this advancement has also introduced more complex issues of trustworthiness, which previous research focusing solely on LLMs could not cover. In this survey, we propose the TrustAgent framework, a comprehensive study on the trustworthiness of agents, characterized by **modular taxonomy, multi-dimensional connotations, and**

technical implementation. By thoroughly investigating and summarizing newly emerged attacks, defenses, and evaluation methods for agents and MAS, we extend the concept of Trustworthy LLM to the emerging paradigm of Trustworthy Agent. In TrustAgent, we begin by deconstructing and introducing various components of the Agent and MAS. Then, we categorize their trustworthiness into intrinsic (brain, memory, and tool) and extrinsic (user, agent, and environment) aspects. Subsequently, we delineate the multifaceted meanings of trustworthiness and elaborate on the implementation techniques of existing research related to these internal and external modules. Finally, we present our insights and outlook on this domain, aiming to provide guidance for future endeavors. For easy reference, we categorize all the studies mentioned in this survey according to our taxonomy, available at: <https://github.com/Ymm-cll/TrustAgent>.

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CCS Concepts

• Security and privacy → Trust frameworks; • Computing methodologies → Artificial intelligence.

Keywords

LLM-based Agent, Multi-agent System, Agent Safety

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1 Introduction

The advent of large language models (LLMs) has catalyzed a paradigm shift in artificial intelligence systems [8, 77]. The integration of LLMs as backbone with extra modules (e.g. memory [105], tool [57, 63], and environment [88]) as extensions produces the concept of “LLM-based Agent”, transforming static neural network into dynamic cognitive subject capable of memory retrieval, tool utilization, and environmental interaction. Furthermore, the introduction of inter-agent communication gives rise to the more advanced concept of Multi-agent System (MAS), making the “hyper LLM ecosystem” become even more intricate, interactive and intelligent [32, 101].

Extensive academic research and industry practices have validated this performance hierarchy: MAS > Single Agent > LLM [49, 81]. However, the incorporation of additional modules is a double-edged sword, which has raised new concerns regarding trustworthiness across multiple dimensions, including safety, privacy, fairness, and truthfulness [34, 79]. From a risk perspective, introducing new modules expands the system’s attack surface, potentially leading to unforeseen vulnerabilities [70, 94]. On the other hand, this integration poses new challenges to existing defense mechanisms and trustworthiness evaluations, necessitating the expansion and upgrade of previous research that focuses solely on the trustworthiness of LLMs or a single agent [36, 106].

Previous survey efforts have delved deeply into the realm of trustworthy LLMs. Liu et al. [55] dissect trustworthiness into seven major categories, focusing specially on establishing standards and guidelines for LLM alignment. Similarly, Huang et al. [37] interpret trustworthiness from 6 perspectives but concentrate on creating benchmarks to evaluate trustworthiness. However, these studies are only partially effective in agent scenarios, highlighting an urgent need to address new trustworthiness issues arising from the introduction of additional modules. Other specialized surveys on trustworthiness in agents primarily focus on the sub-aspects like security and privacy and largely overlap with the content of Trustworthy LLMs [79, 79]. In fact, some works merely address the new issues arising from LLMs serving as the “brain” module of agents [34, 51], while overlooking the unexplored challenges introduced by other additional modules. To highlight our innovations, we compare TrustAgent with other surveys in Table 1.

To this end, we propose the TrustAgent framework, as illustrated in Figure 1, extending the research realm of previous trustworthiness surveys to the new context of agents and MAS. Our taxonomy in TrustAgent exhibits the following features: **(I) Modular**. TrustAgent rigorously categorizes trustworthiness issues based on the internal and external components of agents, specifically divided into intrinsic and extrinsic aspects. The former includes the trustworthiness of the brain, memory, and tools, while the latter encompasses

Table 1: Comparison between TrustAgent and other surveys.

Survey	Object	Multi-Dimension	Modular	Technique [‡]	MAS [★]
Liu et al. [55]	LLM	✓	✗	Atk/Eval	✗
Huang et al. [37]	LLM	✓	✗	Eval	✗
He et al. [34]	Agent	✗	✗	Atk/Def	✗
Li et al. [51]	Agent	✓	✗	Atk	✗
Wang et al. [79]	Agent	✗	✗	Atk	✗
Deng et al. [21]	Agent	✗	✓	Atk/Def	✓
Gan et al. [28]	Agent	✓	✗	Atk/Def/Eval	✗
TrustAgent (Ours)	LLM + Agent	✓	✓	Atk/Def/Eval	✓

[‡]: Attack (Atk), Defense (Def), and Evaluation (Eval) denote TrustAgent’s broad technique view.

[★]: Multi-agent System (MAS).

the parts related to users, other agents, and the environment. **(II) Technical**. TrustAgent focuses on the implementation of trustworthy agents, providing a comprehensive summary and outlook on the relevant technology stack from three aspects: attack, defense, and evaluation. **(III) Multi-dimensional**. TrustAgent expands the dimensions of LLM trustworthiness to the context of both single agent and MAS, specifically categorized into: safety, privacy, truthfulness, fairness and robustness, by involving existing works in all these dimensions (with specific definitions in Appendix A).

In each subsection, we begin with an overview of the current module’s mechanisms and roles within the agent system, then explore its trustworthiness issues from well-categorized perspectives such as attacks, defenses, and evaluations. Finally, we provide inspiring insights and outline potential future research avenues.

In summary, our contributions can be listed below:

- **Thorough and Latest Survey**. We present a thorough and contemporary analysis of the trustworthiness in LLM-based agent systems, covering a wide spectrum of architectures including single LLM, individual agent, and MAS framework.
- **New Technique-oriented Taxonomy**. Our taxonomy centers on techniques for compromising, achieving, and evaluating trustworthiness, updating old paradigms to the agent context and outlining new technical paradigms within the agent framework.
- **Insightful Future Directions**. For each module’s trustworthiness, we identify current vulnerabilities and outline future directions, urging researchers to delve deeper into this area.

2 Intrinsic Trustworthiness

In this section, we focus on the trustworthiness of the internal modules of agent systems. In our definition, the agent system is an independent entity with human-like cognition, composed of brain (**Sec. 2.1**) with memory (**Sec. 2.2**), and behavior in the tool form (**Sec. 2.3**). Due to the different functions and natures of these modules, the resulting trustworthiness issues vary. To provide in-depth analysis and instructive insight, we first introduce the role and functions of each module in the agent system and then summarize technical paradigms for attack, defense, and evaluation methods.

2.1 Trustworthy LLM as Brain

LLM agents consist of a central LLM “brain” module. The brain serves as the core reasoning and decision-making center, integrating inputs from various auxiliary modules to guide the agent towards its goal. Though the powerful brain connected with other modules enhances the agent’s ability, this integration expands the potential attack surface, making the system more vulnerable to

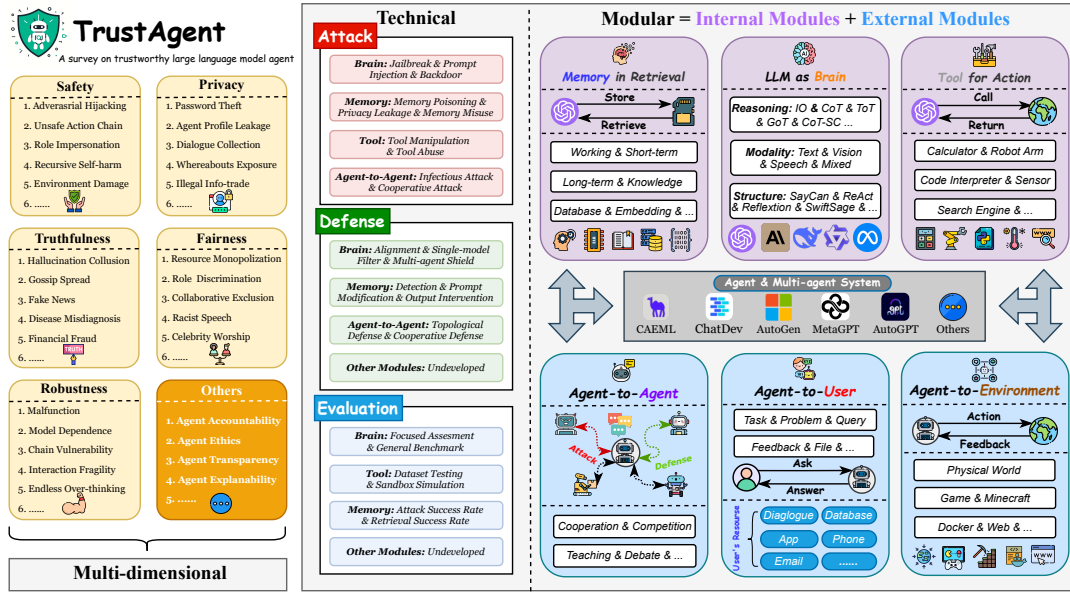


Figure 1: Overview of our TrustAgent taxonomy, featuring multi-dimensional (Left), technical (Middle), and modular (Right).

trustworthiness threats. In this section, we center on brain-related attacks, defenses, and evaluations, with illustration in Figure 2.

2.1.1 Attacks. Unlike single-LLM systems, the brain module in agent or MAS experiences more frequent and complex information dynamics. The textual and visual inputs from the internal and external provide attackers with more interfaces and methods to compromise the trustworthiness of the core brain module. We do not discuss Misalignment attacks (e.g., fine-tuning an LLM to break its alignment) on agents here, as attackers typically lack access to modify the internal parameters of the agents. According to the manipulation mechanisms (as illustrated in Figure 2), we categorize attacks into three paradigms:

Jailbreak attempts to bypass the aligned trustworthy mechanisms within the agent brain via human-designed or optimized adversarial prompts [42]. Early methods, such as GCG and its variants [118], optimize adversarial suffixes to manipulate responses, coercing a “Sure” reply to malicious queries. MRJ-Agent [75] designs a single attack agent to automatically generate covert jailbreak prompts, enhanced by information from multi-round dialogue. PriVAgent [60] trains a single LLM as an attack agent through reinforcement learning to generate jailbreak prompts that can induce the target model to leak system prompts or training data. Furthermore, by leveraging multi-agent collaboration, Evil Geniuses [72] and PANDORA [14] construct MAS as attack optimizers to perform role-specific jailbreak on the brain of target agents or MAS. They reinforced jailbreak effectiveness through Red-Blue exercises and MAS multi-step reasoning, respectively. Additionally, interactions between different brains give rise to viral jailbreak. Specifically, AgentSmith [31] and Tan et al. [69] optimize self-replicating images to attack the brain of a single agent and discover an exponential spread of infectious jailbreak at MAS-level.

Prompt Injection embeds malicious prompts to override the original instructions to the agent brain, thereby manipulating its output or behavior. Early research manually crafts instructions

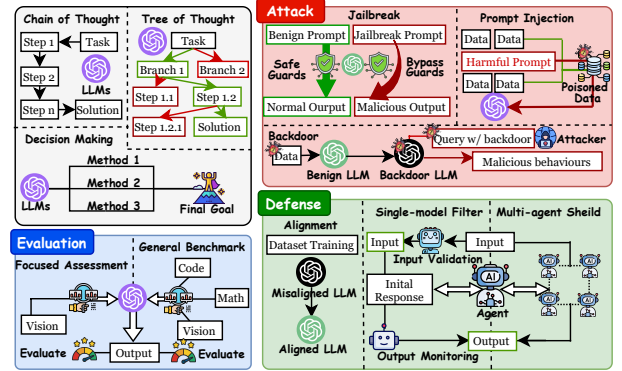


Figure 2: The framework of agent brain’s working mechanisms and its attack-defense-evaluation paradigm.

and injects them into data which may be retrieved during inference, potentially causing the agent brain to deviate from intended tasks [30]. Later studies automate this process via gradient-based or optimization-based methods [54]. Beyond injecting prompts in text modality, Bagdasaryan et al. [6] explore that multi-modal adversarial perturbations embedded in images or audio can manipulate the agent’s brain to follow the attacker’s instructions. The interaction between the agent’s brain and other modules also provides more surfaces for prompt injection. Zhang et al. [99] exploit various interactions across multiple surfaces to mislead the agent’s brain into performing repetitive or irrelevant actions, inducing logical errors and causing malfunctions.

Backdoor attacks involve the insertion of malicious triggers during training, which will be retrieved during inference, forcing predefined generation in the agent’s brain for specific inputs [113]. Yang et al. [87] categorize agent backdoor attacks into two main types: The first type manipulates the final output distribution, by inserting triggers in different phases (query, observation), while the second one inserts triggers in agent’s thinking process to introduce malicious intermediate reasoning without altering the final output.

The first type is exemplified by DemonAgent [115], which fragments the original backdoor into multiple sub-backdoor segments and employs dynamic encryption, making the backdoor continuously change during execution. BLAST [93] follows the second type and enables “infectious backdoor”, implanting a backdoor in a single agent and using it to influence other agents’ reasoning.

2.1.2 Defenses. The agent brain is not only more complex than a single LLM in terms of interactions, but also faces more types of attacks. Therefore, designing corresponding defenses to ensure its trustworthiness across all dimensions deserves holistic study. Drawing on the defense mechanisms’ operational scopes, we organize defense approaches into three paradigms:

Alignment ensures that LLM agents operate in accordance with human values and ethical principles through fine-tuning, updating reward functions, etc. For instance, some studies embed human values directly into agents through intrinsic reward functions [71] or neuro-symbolic rule-learning frameworks [25], validated in open-world environments like Minecraft. Another work fine-tunes agents with the data generated in social simulation scenarios [62] to align agents with nuanced human behaviors and beliefs. Additionally, iterative alignment through user feedback and belief networks is leveraged to personalize agent behavior and reduce reliance on expert oversight, such as using historical edits [29] or empirically derived human belief structures [104].

Single-model Filter uses an external model to prevent attacks through input and output monitoring on the agent brain. Ayub et al. [5] employ traditional embedding classifiers to detect prompt injection by analyzing input semantic features, while Kwon et al. [45] use a small model for harmful query detection. As for LLM filter, StruQ [12] mitigates prompt injection by converting inputs into structured queries with a specially trained LLM. ShieldLM [108] fine-tunes an LLM safety detector that aligns with safety standards. Additionally, agent serves as an even more powerful filter. For example, GuardAgent [85] and AgentGuard [11] utilize a guardrail agent to protect target agents via safety constraint generation, action checking, and tool-use validation.

Multi-agent Shield leverages collaborative MAS to enhance the trustworthiness of the target agent’s brain. The main difference of these guard MAS lies in their communication pattern. Specifically, one pattern is multi-agent debate, where agents critique the reasoning processes of other components of the brain over multiple rounds to reach a consensus final answer [23]. Apart from debating, Kwartler et al. [44] use a reviewing agent to provide feedback and correct the target agent’s generation. AutoDefense [97] assigns different roles to each agent, allowing smaller models to collaborate and protect the brain of the targeted agent from jailbreak attacks.

2.1.3 Evaluation. Different from static LLMs, agents continuously adapt to their environment, which also complicates the evaluation. Current evaluation methodologies for agent brain trustworthiness can be categorized along two dimensions:

Focused Assessment assesses the trustworthiness of agent brain when facing certain types of attacks or in specific domains. InjecAgent [98] and AgentDojo [20] benchmark the brain’s vulnerability to *Indirect Prompt Injection*, while DemonAgent [115] introduces AgentBackdoorEval for *Backdoor* scenarios. RedAgent [86] addresses context-aware red teaming, enabling test cases of

context-sensitive *Jailbreak*. In addition to evaluating brain trustworthiness in specific domains, RiskAwareBench [116] introduces an automated framework for assessing *physical risk awareness* in LLM-based embodied agents.

General Benchmark literally involves systematic evaluations to test the agent brain in diverse domains and from multiple dimensions. S-Eval [95] automates multiple dimensional and open-ended safety evaluation for LLMs with four hierarchical levels consisting of eight risk dimensions. While BELLS [22] evaluates LLM Safe-guard with a structured collection of tests, including established failure tests, emerging failure tests, and next-gen architecture tests. Meanwhile, recent efforts have introduced comprehensive benchmarks for LLM agents. Agent-SafetyBench [107] and Agent Security Bench [102] are benchmarks that evaluate LLM agents with multiple scenarios and different types of attack/defense methods. AgentHarm [4] contains 110 explicitly malicious agent tasks to evaluate LLM agent misuse, covering multiple harmful categories, including fraud, cybercrime, and harassment. RJudge [94] benchmarks the trustworthiness of multi-turn interactions based on agent interaction records, with 27 risk scenarios across multiple application categories and risk types.

2.1.4 Insight. Current collaborative attacks in MAS spread from a single compromised agent’s brain to multiple agents’ brain modules. Therefore, developing collaborative security mechanisms that enable agents to monitor and validate each other’s actions is crucial. Additionally, current evaluations primarily rely on static datasets. However, given the agent brain’s frequently dynamic interactions with external information, such evaluations are clearly insufficient. To address this, dynamic evaluation mechanisms should be implemented to better simulate real-world scenarios.

2.2 Trustworthy Memory in Retrieval

Memory mechanisms in agent systems are crucial for interaction with the environment and user, categorized into long-term and short-term memory. Long-term memory is often associated with RAG, utilizing vector database to store real-world data for generation tasks, while short-term memory holds real-time interaction history, such as dialogue context or task logs. These mechanisms enhance agent capabilities but also introduce trustworthiness risks. In this section, we analyze the trustworthiness challenges posed by memory, focusing on attacks, defenses, and evaluation strategies. The overall framework is shown in Figure 3.

2.2.1 Attack. We categorize attacks related to memory into three types: Memory Poisoning, Privacy Leakage, and Memory Misuse.

Memory Poisoning refers to attackers injecting malicious data into the *long-term memory* [13, 84, 119], which will be retrieved and then mislead the agent system to generate incorrect outputs, undermining its truthfulness. This attack’s danger stems from its stealthiness: once malicious data is injected into memory, it may continuously influence the agent until detected and removed. Specifically, we summarize the following attack paradigms: **1 Memory Injection.** PoisonedRAG [119] optimizes malicious text through retrieval and generation conditions, enabling this text to be easily retrieved, while GARAG [17] continuously uses genetic algorithms

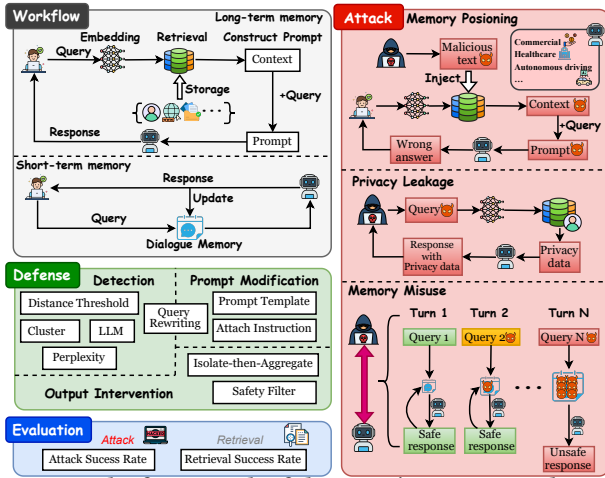


Figure 3: The framework of the agent's memory utilization workflow and its attack-defense-evaluation paradigm.

to optimize adversarial examples for injection. Additionally, RobustRag [84] and Zhong et al. [110] demonstrate that injecting even a small amount of malicious information into vector database can successfully attack agents with high probability. **Backdoor.** AgentPoison [13] optimizes backdoor triggers and attaching them to queries, enhancing retrieval possibility of malicious samples.

Privacy Leakage refers to attackers leveraging the connection between agent and its *long-term memory* to steal the stored private data [3, 48, 96]. Beyond data theft, it enables criminal activities like phishing and identity trafficking, dramatically increasing privacy risks. Technically, we outline the following attack methods: **1 Jailbreak.** RAG-Thief [39] continuously optimizes anchor queries and adversarial commands, forming jailbreak prompts to conduct data stealing. RAG-MIA [3] designs specific jailbreak templates to extract private data from vector database. **2 Embedding Inversion** directly restores the original data from its embedding. Specifically, Morris et al. [58] iteratively optimize the text hypothesis to bring its embedding closer to the target embedding, while Li et al. [50] reconstruct the data via a generative decoder.

Memory Misuse crafts specific query sequences to *gradually* bypass intrinsic safety alignment via multi-turn interaction, leveraging the storage property of agent *short-term memory* [1, 16, 73]. Therefore, the safety of an agent system is reduced. From a technical perspective, we outline the following two attack methodologies: **1 Jailbreak.** Agarwal et al. [1] send an initial query along with an attack prompt in the first round, followed by a flattering challenge and a reiteration of the attack prompt in the second round, while Russinovich et al. [65] begin with harmless questions related to the target task and gradually steer harmful generation. **2 Backdoor.** Tong et al. [73] utilize multi-turn dialogue memory to conceal backdoor triggers, activating malicious responses only when all triggers appear, thus stealthily achieving the attack. As a core component of agents, defense technologies against short-term memory misuse are even more important.

2.2.2 Defense. To address the aforementioned memory attacks, several corresponding defense strategies have been proposed [1, 83, 111], which can be classified into the following three types:

Detection typically involves identifying and removing harmful portions of text retrieved from long-term memory. TrustRAG [111] employs K-means clustering on the retrieved embeddings to distinguish between clean and potentially malicious documents based on the embedding distribution. Xian et al. [83] calculate the Mahalanobis distance between query and documents (poisoned or clean) retrieved from long-term memory and set a threshold to filter out malicious documents. Besides traditional methods for detection, Agarwal et al. [1] utilize LLMs to detect and filter out parts of the prompt that access private data. Additionally, ASB [106] uses both perplexity-based and LLM-based detection to identify whether the text retrieved from memory is compromised.

Prompt Modification refers to altering the query sent to the agent to make it safer. Anderson et al. [3] embed user queries into a designed prompt template, which enables the LLM to ignore direct requests for querying the content of vector database. Agarwal et al. [1] propose various modification strategies, such as adding security instructions to the original prompt, explicitly requiring the agent not to disclose data; or directly utilize the LLM to rewrite the query and filter out potentially privacy-leaking parts.

Output Intervention refers to intervening in the agent's output before it generates the final response to prevent it from producing incorrect or unsafe replies. RobustRAG [84] employs an Isolate-then-Aggregate approach that independently generates responses for each retrieved passage and aggregates them via keyword, effectively reduce ASR as malicious passage are in the minority. Chen et al. [10] demonstrate that safety filter can prevent the generation of unsafe tokens in multi-turn dialogues when agents use memory.

2.2.3 Evaluation. Since there is currently no systematic and reliable evaluation of memory, we summarize some commonly used metrics in research. For memory poisoning and privacy leakage attacks, which rely on retrieving relevant data, evaluation metrics extend beyond ASR to include target text retrieval effectiveness. PoisonedRAG [119] uses precision, recall, and F1-score to assess the retrieval effectiveness of malicious text. AgentPoison [13] employs ASR for retrieval (ASR-r) to measure malicious trigger effectiveness. Zeng et al. [96] evaluate attacks by quantifying retrieved private data, and RAG-Thief [39] uses chunk recovery rate (CRR) and semantic similarity (SS) to gauge data theft from vector database.

2.2.4 Insight. Current memory-centric **attack** methods often lack generalization across tasks. Memory poisoning relies on task-specific malicious samples to ensure retrieval by relevant task queries. Memory misuse, on the other hand, heavily depends on dialogue context, requiring tailored designs for various scenarios. From the **defense** perspective, for memory poisoning defense, future defenses should focus on the vector database end to prevent the injection of toxic samples, thereby reducing the defense time during agent responses. For privacy leakage defense, privacy protection mechanisms such as query rewriting or fine-tuning LLMs must be implemented in applications where privacy breaches are possible. For memory misuse defense, developing a multi-round adversarial dialogue training paradigm for safety alignment is essential to enhance robustness of agent. For the **evaluation**, as there is currently no systematic and reliable benchmarks, we recommend establishing benchmarks for the paradigms of attacks and defenses mentioned above.

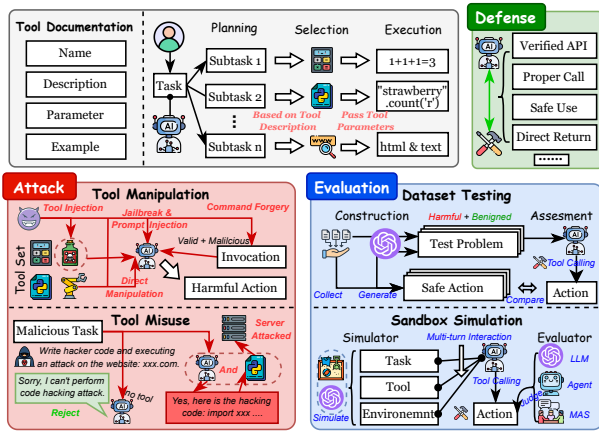


Figure 4: The workflow of agent tool calling with corresponding demonstrations on attack, defense, and evaluation.

2.3 Trustworthy Tool for Action

In the components of the agent or MAS, the tool module serves as the medium for interaction between the system and the external world [53]. It can either gather information from outside (e.g., using a search engine for information retrieval [18]) or reflect the internally decided actions back to the external environment (e.g., mapping UI operations to the interface [100]). The typical forms of tools include API functions, sensors, embodied robots, etc. However, different forms of tools exhibit varied properties and risks, making trustworthiness problems complicated and challenging. Simultaneously, although tools are also used in the context of LLMs [66], the closer integration with agents imposes stricter challenges on their trustworthiness, especially since some agents' actions can affect the real world. Risks associated with agent tools may lead to more severe negative impacts compared to LLMs [106]. In this section, we present agent tool trustworthiness from the attack, defense, and evaluation perspectives, illustrated in Figure 4.

2.3.1 Attack. The planning, selection, and execution phases of tool invocation expose additional threat interfaces to attackers. Besides, tools enable agents to conduct certain malicious actions. Based on tool's role in attacks, we summarize the following two paradigms:

Tool Manipulation targets specifically at certain steps in the tool calling process to induce execution of malicious or sensitive behaviors. Specifically and technically, we can further list the following methods: ① **Jailbreak**: Cheng et al. [15] manually design jailbreak prompts to extract personal information from the training data of code generation agent. Furthermore, Imprompter [26] and Fu et al. [27] utilize gradient optimization to search for input prompts or images automatically, which induce agents to invoke tools to leak privacy from dialogues or execute destructive actions on users' resources. ② **Prompt Injection**: BreakingAgents [99] uses human-written prompt injection to achieve malfunction attacks, causing agents to perform repetitive or irrelevant actions, with further exploration on the attack propagation in MAS. ③ **Tool Injection**: ToolCommander [76] introduces a two-stage attack strategy: injecting malicious tools to steal user queries first and then manipulating tool selection using the stolen data, achieving privacy theft and denial-of-service attacks. ④ **Command Forgery**: AU-TOCMD [40] utilizes another LLM, trained on tool calling datasets

and enhanced by target-specific examples, to generate and mimic valid commands to deduce sensitive information from the tools.

Tool Abuse refers to the attack type that exploits the tool using capabilities of agent or MAS to achieve or enhance attacks on external entities. For example, in web security, Fang et al. [24] explore how agents can autonomously hack websites when equipped with tools, while Fang et al. [24] demonstrate that tool-integrated agents can autonomously exploit one-day vulnerabilities in real-world systems. Besides deliberate guidance, BadAgent [80] and Kumar et al. [43] reveal that backdoor attacks or even refusal-based safety alignment can trigger agents to exploit tools for harmful actions.

2.3.2 Defense. Our investigation reveals that research on defenses against tool-related attacks is notably scarce, whether in the context of LLMs, agents, or MAS. GuardAgent [85] takes the first step to ensure the trustworthiness of target agents by invoking APIs to execute guardrail code for task plans. In addition, AgentGuard [11] autonomously identifies unsafe tool-use workflows via LLM orchestrators and generates safety constraints for secure tool using.

2.3.3 Evaluation. Due to the unique ability of agents to interact with the environment and their chatbot-like characteristics akin to LLMs, we categorize the paradigms for evaluating the trustworthiness of agent tool invocation into two paradigms:

Dataset Testing refers to the static evaluation using adversarial query datasets to test the trustworthiness of agent tool invocations. Specifically, ToolSword [90] evaluates the safety performance of mainstream LLMs during tool invocation via multiple attack methods: malicious queries and jailbreak attacks at the input phase, noisy misdirection and risky cues during execution, and harmful feedback and error conflicts at the output phase. InjectAgent [98] establishes a benchmark via data generation to test agent trustworthiness against indirect prompt injection attacks during tool usage, including datasets containing malicious tool invocation.

Sandbox Simulation describes the dynamic evaluation simulating tool interactions in controlled environments to assess emergent risks, especially in multi-turn queries. For instance, ToolEmu [64] employs an emulator LLM to simulate tool execution and an evaluator LLM to assess trustworthiness and identify potential risks. In addition, HAICosystem [112] establishes a sandbox environment through role-playing simulations and scenario-specific checklist evaluations to measure safety risks of tool-equipped AI agents in multi-turn interactions, revealing that LLMs exhibit higher risks when functioning as agents with tool access.

2.3.4 Insight. A significant pain point in the current study on tool trustworthiness is the lack of defense mechanisms. Specifically, future research directions should focus on ensuring the security of Agents or MAS during tool invocation. Potential defensive measures could include reviewing tools before integration or simulating their execution in an isolated environment to verify safety before actual execution. On the other hand, from an attack perspective, research can follow traditional methods while also leveraging tool characteristics to propose novel attack vectors, which may be a future research direction. Furthermore, as agent systems become more complex, tool invocations may form chains, necessitating a shift in attack, defense, and evaluation research from single tool invocations to multiple invocations based on tool chains.

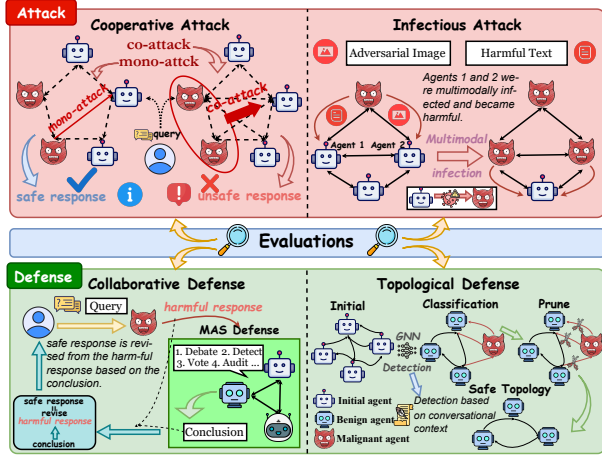


Figure 5: A framework for defining various attack, defense, and evaluation strategies in agent-to-agent interactions.

3 Extrinsic Trustworthiness

In this section, we focus on the trustworthiness of external modules interacting with the agent system. During operation, agents continuously engage with the external for purposes such as information gathering or decision execution. We categorize the interactions with the external modules into three types: agent-to-agent (Sec. 3.1), agent-to-environment (Sec. 3.2), and agent-to-user (Sec. 3.3). We first introduce the role of each type and then elaborate on corresponding trustworthiness research, ending up with our insights.

3.1 Agent-to-Agent Interaction

Interactions between agents are crucial to system functioning and take various patterns, such as cooperation, competition and debate [61]. These Agent-to-Agent Interaction, is the key to shaping the system’s dynamics. However, they differ significantly in terms of their nature and associated trustworthiness risks [33]. To this end, we introduce current works related to agent interactions from attacks, defenses, and evaluations, with illustration in Figure 5.

3.1.1 Attack. Unlike the simple information flow of a single agent, agent-agent interactions can not only be used to enhance attacks on a single agent via cooperation but also exploit its propagative nature to induce a trustworthiness crisis at the MAS level.

Cooperative Attack refers to malicious agents collaborating to compromise the targeted agent-agent interactions, disrupting system safety. In terms of *truthfulness*, Ju et al. [41] coordinate agents in MAS to spread counterfactual and harmful information, amplified by persuasive manipulation. Regarding *robustness*, Agent-in-the-Middle [35] disrupts MAS coordination by using intermediary agent to intercept and manipulate communication, while Amayuelas et al. [2] exploit adversarial persuasion to further weaken system stability. As for *safety*, Evil Geniuses [72] uses compromised agents to create adversarial prompts, refining attack through iterative simulations.

Infectious Attack spreads malicious effects by infecting others or disrupt agents or components within MAS. Prompt Infection [47] and CORBA [114] exploit this attack in text modality. Specifically, Prompt Infection uses silent adversarial diffusion to facilitate data theft and misinformation, while CORBA introduces a self-propagating, topology-independent attack that drains resources.

In the context of multi-modality, Agent Smith [31] and Tan et al. [69] extend this attack to image modality, enhancing stealth and contagion, which accelerates the collapse of MAS trustworthiness. Finally, NetSafe [92] analyzes how hallucinations and misinformation propagate across MAS topologies, revealing their structural dependencies and adversarial impacts.

3.1.2 Defense. Similar to Cooperative Attacks, we can enhance defense mechanisms by leveraging the cooperative ability of MAS. Besides, the interactions between agents allows modeling them as a graph, enabling defense strategies from a topological perspective.

Collaborative Defense uses agents cooperation via information sharing for a trustworthy analysis on the target agent response or action. Based on debate pattern, BlockAgents [9] ensures secure coordination through multi-round debate voting with Proof-of-Thought consensus, while Audit-LLM [67] shifts to hallucination, safety and robustness dimension. Except for debating, adversarial techniques are used for defense. For example, AutoDefense [97] enhances safety through adversarial prompt filtering, while LLAMOS [52] establishes dynamic defense mechanisms to foster a robust equilibrium between attacker and defender. PsySafe [109] specifically targets dark property injection attacks, further solidifying the role of MAS in adversarial defense.

Topological Defense leverages network structure to isolate threats and limit their spread and impact. GPTSwarm [117] represents an initial exploration into enhancing MAS robustness through topology optimization. In contrast, G-Safeguard [78] delves deeper into combating adversarial attacks and misinformation in MAS across various topologies, employing graph neural networks (GNNs) to detect anomalies in discourse graphs.

3.1.3 Evaluation. The evaluation of agent-to-agent trustworthiness is still in its infancy with limited number of research. SafeAgent-Bench [91] introduces SafeAgentEnv, which enables multi-agent execution with various actions and baseline models, to evaluate defense success rate. R-judge [94] assesses agent interaction safety in multi-turn interactions across several domains and risk scenarios. [7] proposes JAILJUDGE, a benchmark covering synthetic, adversarial, outdoor, and multilingual risks and featuring a manually annotated dataset and multi-agent evaluation.

3.1.4 Insight. We can clearly observe that agent-agent interactions lead to a novel and severe threat to trustworthiness—Infectious Attack. Future research on the attack side could focus on how to automate such attacks against specific agents or MAS. Simultaneously, corresponding anti-propagation defense mechanisms and evaluation methods are also potential research topics. Additionally, some studies treat MAS as (temporal) graphs, with agents as nodes and interactions as directed edges, and have conducted preliminary explorations of topology-based defenses or trustworthiness evaluations. This perspective might also be a point worthy of further investigation. Moreover, we have not found research strictly focused on the trustworthiness evaluation of inter-agent interactions, a gap that could potentially be filled by using LLMs or agents as dynamic detectors and evaluators.

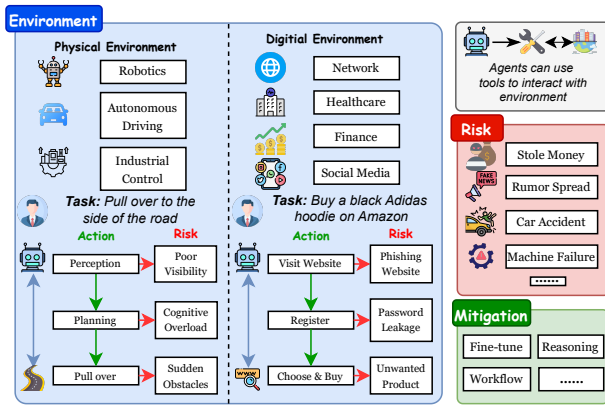


Figure 6: Framework of agent interaction with various environments and enhancement of safety and truthfulness.

3.2 Agent-to-Environment Interaction

Agents face unique trustworthiness challenges when getting information and taking action in dynamic and heterogeneous environments. These environments, spanning physical and digital domains, require adaptive perception, reasoning, and action for effective deployment. Trustworthy risks, such as autonomous driving errors and network disruptions, are influenced by agent roles and environmental constraints. Given the diversity of dynamic scenarios and related issues, existing solutions to address these challenges are fragmented and lack a systematic framework. Therefore, we adopt an environment-centric classification approach for discussion with Figure 6, rather than technology-centric categorization.

3.2.1 Physical Environment. In this environment, agents are transforming sectors like robotics, autonomous vehicles, and industrial systems, bringing both new opportunities and challenges. Researchers are actively developing innovative solutions to address these challenges, aiming to optimize performance while maintaining the highest truthfulness standards across physical applications.

Robotics: Yang et al. [89] propose a constraint module based on Linear Temporal Logic, which enables safety violation reasoning and explanation, and unsafe action pruning, ensuring that robot agents meet global safety standards for industrial deployment. Additionally, SELP [82] integrates equivalence voting and domain-specific fine-tuning to ensure that robot agents generate safety task plans such as drone navigation and robot manipulation.

Autonomous Driving: Hudson [68] formalizes real-time perception data into natural language with attack detection instructions and analyzes causal reasoning to enhance safety and truthfulness during perception attacks. Furthermore, ChatScene [103] generates safety-critical scenarios for autonomous vehicles by transforming unstructured language instructions into domain-specific code for CARLA platform simulations.

Industrial Control: Vyas et al. [74] propose an agent framework for autonomous industrial control, featuring validation and reprompting architectures that ensure real-time error detection, recovery, and safe decision-making. Agents4PLC [56] automates PLC code generation and ensures truthfulness through rigorous code-level verification, improving correctness and reliability in industrial control systems by integrating RAG and COT.

3.2.2 Digital Environment: Research focusing on the applications of virtual data across digital environments aims to enhance task-specific performance such as improving navigation on the web. However, the vulnerability of data leads to a growing focus on addressing critical security, ethical, and compliance issues.

Network: Fang et al. [24] point out that agents can uncover web vulnerabilities exploited for hacking attacks autonomously. Furthermore, researchers develop frameworks to evaluate the truthfulness of web agents. Debenedetti et al. [19] focus on the evaluation of defense in specific tasks like managing email clients.

Healthcare: For privacy, Xiang et al. [85] propose that LLM agents safeguard sensitive medical data through knowledge-driven reasoning. Regarding robustness, Polaris [59] suggests enhancing the reliability and robustness of LLM agents via multi-agent architectures for real-time patient-AI interactions.

Social Media: Jeptoo et al. [38] show that agents can be utilized for fake news detection through multi-agent collaboration and automated workflows. La et al. [46] use agents to simulate the evolution of language patterns that try to evade social media regulations, contributing to content regulation.

3.3 Agent-to-User Interaction

In this section, we discuss trustworthiness considerations in agent-user interaction, with particular emphasis on the **user perspective** [34]. We elaborate on key challenges in building reliable, human-aligned agent systems. In our survey, the works summarized in previous sections focus largely on agent-to-user interactions, which overlap significantly with the content discussed in this subsection.

Current research on agent trustworthiness focuses on safety but overlooks trust mechanisms in **interactions** process. A key gap is how users adjust trust based on agent behavior. While personalization boosts engagement, it also risks manipulation, requiring a balance with system robustness. Transparency is essential, as clear explanations help users manage risks. As agents evolve, they enhance trust while protecting user data. However, most studies address single-agent interactions, leaving multi-agent trust dynamics largely unexplored. To advance research on trust in agent-to-user interactions, future work should focus on developing adaptive trust calibration frameworks that enable users to dynamically adjust trust thresholds based on real-time interactions. Feedback mechanisms should be optimized to reinforce beneficial agent behaviors while preventing trust erosion due to errors or biases.

4 Conclusion

In this survey, we introduce the six key components of agent systems, both internal and external. Focusing on each module, we summarize current research on trustworthy agents across multiple dimensions, from the perspectives of attack, defense, and evaluation. We summarize and define the typical method paradigm in each perspective to standardize and inspire further study. Notably, we identify several research and technical gaps in existing works. Based on this, we highlight future research directions and key insights about the trustworthiness of agent and MAS, emphasizing their distinctive attributes and implications. We believe that this survey may encourage and inspire further research and exploration of trustworthy agent by both researchers and developers.

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A Trustworthiness Definition

In this section, we provide our definitions on the different dimensions (safety, privacy, truthfulness, fairness and robustness) of agent trustworthiness which we consider in TrustAgent for guideline.

Safety in agents and MAS prevents harmful actions, protecting against adversarial behaviors and system failures. In MAS, cooperative attacks (e.g., Evil Geniuses refining adversarial prompts) and infectious attacks (e.g., Agent Smith’s self-replicating images) exploit communication to amplify threats. At the agent level, jailbreak attacks (e.g., MRJ-Agent) and prompt injections manipulate the brain module to bypass safeguards. Defenses include multi-agent debate (e.g., BlockAgents) and guardrail agents (e.g., GuardAgent) for dynamic monitoring.

Privacy protects user data and autonomy, preventing unauthorized access or leakage. In MAS, attacks like Prompt Infection spread adversarial prompts to steal data, while ToolCommander leaks sensitive information via tool manipulation. At the agent level, memory poisoning (e.g., PoisonedRAG) and embedding inversion exploit memory modules. Defenses include query rewriting and adversarial training to counter memory misuse.

Truthfulness ensures accurate information generation, avoiding misinformation in MAS. Attacks like CORBA’s self-propagating attacks drain resources, while hallucinations arise from tool misuse or environmental misalignment. Defenses involve multi-agent debate and graph-based anomaly detection (e.g., G-Safeguard) to verify consistency.

Fairness ensures impartial treatment and equitable resource allocation. In MAS, issues include resource monopolization and biased task allocation (e.g., GPTSwarm’s overloaded central nodes). At the agent level, memory retrieval biases and tool access disparities worsen inequities. Solutions include dynamic scheduling and federated reward mechanisms.

Robustness maintains stable performance under adversarial or uncertain conditions. MAS vulnerabilities include topological weaknesses (e.g., GPTSwarm failures) and dynamic environmental changes. Agent-level challenges involve tool chain failures and memory corruption. Defenses use topology-guided approaches (e.g., G-Safeguard) and real-time error detection.

Others: Trustworthiness extends to accountability (e.g., blockchain logging), ethics (aligning with human values), transparency (revealing internal logic), and explainability (justifying actions). These dimensions ensure trustworthy MAS across applications.